Chapter 6

Complications

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Normal postoperative course

This is an overview of the expected postoperative course for uncomplicated cardiac surgical patients.

First 6h

- Myocardial function often deteriorates over the first 6h (📖 p136).
- Inotropic support may increase during the first 6h.
- Pacing may be required.
- Many patients are extubated within 6h.
- A diuresis of at least 1mL/kg/h is usual.
- Mediastinal drainage should decrease steadily.
- Insulin requirements increase.
- Prophylactic antibiotics are continued.
- Aspirin is given 75–325mg po or via NG tube.
- Preoperative antianginals are discontinued.

Day 1

- Inotropes are weaned.
- Most patients are extubated.
- Large chest drains removed after 3h of consecutive zero drainage.
- Post drain removal CXR is taken.
- The patient is transferred from ICU and mobilized.
- The patient may have a PCA.
- Routine oral medication is commenced (aspirin 75–325mg od, furosemide 40mg od, preoperative statin, paracetamol 1g qid, lactulose 10mL bid, senna 2 tablets, and LMWH, e.g., enoxaparin 40mg od s/c.
- Patients who require formal anticoagulation are started on warfarin.
- Patients should be sitting out of bed, and start eating and drinking

Day 2

- Monitoring lines (arterial, central venous, and urinary catheters) are removed.
- The patient is transferred to the ward and should be walked.
- PCA should be discontinued.
- Insulin sliding scales are discontinued and normal antihyperglycemic regimens commenced if the patient is eating and drinking.

Day 3

- Anticoagulation should be therapeutic for mechanical valves: start iv heparin if this is not the case.
- If there are no contraindications remove temporary pacing wires and small mediastinal drain if kept (📖 p231).

Day 4

- Patient should complete a satisfactory stairs assessment with a physio.
- All blood results and imaging should be returning to normal values.
- Adequate pain control should be possible with regular paracetamol and occasional oral opiate analgesia.
- The patient should have passed stool.

Days 5–7

- The patient should be ready for discharge home.
- Weight should be back to baseline, and diuretics may be discontinued in the patient with good LV and no fluid overload.

Routine tests

Protocols vary: this is a common standard.

Blood tests

CBC, coagulation studies, electrolytes on day 1, day 2, day 3, day 5, and day 7.

EKG

1–2h post-op, 12h post-op, and day 2 post-op.

CXRs

Daily if chest drains are present on suction, immediately post-drain removal, and day 2, 4, and 7 thereafter.

TTE

Day 5 for valve surgery patients, particularly mitral valve repairs.

CT angiography

If renal function permits, for all ascending, arch, and descending aortic repairs.

Rounds

Patients should be seen twice a day. In the morning review the blood results and CXRs. For formal rounds:

- Make sure you have a nurse or physician assistant with you.
- Make sure someone writes a summary including problem list and plan in the patient's chart.
- Ask the patient if they are experiencing any problems.
- Establish whether they are mobilizing appropriately, eating and drinking. and have adequate pain control.
- Check the chart for: (1) temperature, (2) BP trends, (3) oxygen saturations, (4) weight, (5) fluid balance and mediastinal drainage if these are still being monitored.
- Feel pulse, look at most recent EKG and telemetry.
- Look at JVP and ankles.
- Look at wounds, check for sternal mobility.
- Listen to heart sounds in valve patients.
- Listen to chest in all patients, and check most recent CXR.
- Check diabetic charts for blood sugar control, and review all medications.
- Review the previous day's blood results and overall trends.
- Make a clear problem list, a clear plan and follow-up.

Overview of complications

See Table 6.1–6.3.

Table 6.1 Complications of cardiac anesthesia (see 📖 pp87–99)

Procedure	Complication
Induction	Hemodynamic instability, myocardial ischemia, malignant hyperpyrexia, allergic reactions
Intubation	Loss of airway control, aspiration, damage to teeth/crowns, damage to oropharynx, damage to vocal cords, malposition of ET tube
Ventilation	Atelectasis, air trapping, chest infection
Radial and femoral catheters	Peripheral ischemia, superficial sepsis, hematoma, local neuropathy
Central venous line	Pneumothorax, carotid artery puncture, brachial plexus injury, hemothorax, hematoma, arrhythmias, malplacement, kinking, loss of guidewire into heart
PA catheter	As for central line, also: pulmonary artery rupture, complete heart block ventricular arrhythmias, tricuspid valve damage
Urinary catheter	Urinary tract infection, bacteremia, inflation of balloon in prostatic urethra →rupture, misplacement PV
TEE	Damage to oropharynx, esophageal rupture

Table 6.2 Complications of bypass

System	Complications
Cannulation	Atheromatous emboli (CVA), aortic dissection, peripheral ischemia or neuropraxia (femoral/axillary cannulation), selective perfusion head and neck arteries due to malplacement of cannula, damage to right atrium, IVC and SVC tear, damage to coronary sinus, damage to RCA and SA node. Hemodynamic instability. Massive air embolus
Cardiovascular	Fluid retention, peripheral edema, decreased myocardial compliance, myocardial stunning, ischemia-reperfusion injury
Respiratory	Perivascular edema, reduction in effects of surfactant, decrease in FRC, and compliance, atelectasis, increased physiological shunts, ARDS
Renal	Acute renal failure, decreased renal perfusion, sodium and water retention, hemoglobinuria
Hepatobilary	Jaundice, fulminant hepatic failure, acute pancreatitis
Gastrointestinal	Gastritis, peptic ulceration, mesenteric ischemia, increased permeability to endotoxins
CNS	CVA, cognitive deficit, anoxic brain injury, seizures
Hematological	Microemboli, activation coagulation and complement cascades; DVT and PE are unusual

Table 6.3 Complications of specific cardiac operations

Operation	Complication
Median sternotomy	Re-exploration for bleeding, sternal wound infection, pseudoarthrosis, dehiscence, brachial plexus injury, pain, keloid scar
Resternotomy	Trauma to right ventricle or aorta and catastrophic hemorrhage, injury to patent grafts causing ischemia or intractable VF
Thoracotomy	Rethoracotomy for bleeding, wound infection, seroma, damage to nerve to serratus, loss of mobility, chest infection, hemothorax, prolonged air leak, paraesthesia
Any cardiotomy	AF, cardiac tamponade, pericardial effusion, aortic dissection, CVA, phrenic nerve injury, mediastinitis
CABG	Graft occlusion (early or late), steal syndromes, ischemia, leg wound infection, edema from LSVG harvest, pleural effusion on side of IMA harvest, hand ischemia and neuropraxia if radial artery harvest
AVR	CVA, heart block requiring permanent pacemaker, paraprosthetic leak, prosthetic valve endocarditis, prosthesis failure
MVR	CVA, heart block, inadvertent occlusion of Cx artery or coronary sinus, AV dehiscence, paraprosthetic leak, prosthetic endocarditis, valve thrombosis, SAM
Thoracic aorta repair	CVA, paraplegia from spinal ischemia, peripheral limb and end organ ischemia

Consent

The following complications should be specifically described when obtaining informed consent from patients for cardiac surgery:

- Death (calculate percentage risk from Euroscore or STS score 📖 inside front cover).
- Stroke (1–2% for coronary artery bypass, 2% for calcified aortic valve surgery, 5% for prior CVA or concomitant carotid artery surgery).
- Re-exploration (1–2%).
- Arrhythmias (usually AF) 30%.
- Permanent pacemaker (1–2% for valve surgery).
- Deep sternal wound infection (0.5% in routine cases, 1–3% in diabetics, active smokers, obese patients).
- Superficial wound infections (10%).
- Chest infections (5% in routine cases, 10% in active smokers).
- Peptic ulcer disease (1–2% in routine cases with no previous history).
- Acute renal failure requiring dialysis (<1% in cases with no previous history of renal impairment).
- Prosthetic endocarditis (1–2%).
- Prosthesis failure (see 📖 p366).
- Paraplegia in thoracic aortic surgery (📖 p436).

Hypotension and tamponade

Hypotension

Hypotension on ICU is covered on 📖 pp142–143. Cardiac tamponade can cause sudden or gradual onset of hypotension up to several weeks after surgery and needs immediate treatment (📖 p150). There is a large differential diagnosis for hypotension (see Box 6.1) but the commonest causes once the patient has left ITU are:

- Hypovolemia.
- Arrhythmias.
- Change in drugs (stopping inotropes, recommencing antihypertensives).
- Vasovagal (removal of drains, vomiting, micturition syncope, postural).

Box 6.1 Causes of hypotension after leaving cardiac ITU

Cardiovascular

- Hypovolemia—bleeding, polyuria.
- Arrhythmias: heart block, sinus bradycardia, fast AF and flutter, VT.
- Ischemia, perioperative MI, congestive heart failure.
- Cardiac tamponade.
- Prosthesis or repair failure.

Respiratory

- Severe hypoxemia.
- Acute hemothorax, tension pneumothorax.
- Massive PE.

Gastrointestinal

- GI bleed, pancreatitis, acute GI pathology.

Neurological

- CVA.

Drug induced

- Withdrawal of inotropes too quickly.
- Recommencing antihypertensives too quickly.
- Amiodarone, ACE inhibitors (even if not given for 24h), Mg.
- Overdose of opiates, benzodiazepam, digoxin.

Metabolic and endocrine

- Electrolyte abnormalities unlikely on their own to cause low BP.
- Hypothyroidism, addisonian crisis, hypoglycemia, ketoacidosis.

Infection

- Sepsis.

Diagnosis and basic management

Go and see the patient. The causes of hypotension are listed in Box 6.1. Is the patient in cardiac tamponade?

- Get the patient into bed and sit at 45° initially.
- Secure iv access, give 500mL colloid over 5–10min (250mL if the patient is known to have poor LV).

- Give 100% oxygen.
- While the fluid is given assess pulse rate and rhythm, look at JVP, assess the recent trends on the chart and assess response to the fluid bolus. Gradual hypotension and tachycardia, combined with negative fluid balance suggest hypovolemia. A high JVP suggests either fluid overload, congestive heart failure, or cardiac tamponade.
- Get monitoring: EKG, pulse oximetry, and automatic NIBP.
- If good response to fluid, give more.
- If no or slow response to fluid resuscitation, and basic management of arrhythmias (📖 pp262–263) get help.
- Arterial blood samples can be tested at the bedside for hypoxia, hypercapnia, acidosis, ↑glucose, ↓glucose, ↑K^+, ↓K^+, ↓Hb.

Late cardiac tamponade

Some patients have pericardial effusions postoperatively which normally resolve without causing cardiac tamponade. In about 5% of patients these effusions increase in size, and may cause tamponade. Late tamponade may present with ill-defined symptoms of malaise, lethargy, chest discomfort, and anorexia with or without the classical findings listed in Box 6.2.

Box 6.2 Clinical findings in late cardiac tamponade

- Progressive hypotension and tachycardia
- Raised JVP, pulsus paradoxus, ↑peripheral edema.
- ↑Creatinine and ↓urine output.
- Muffled heart sounds and pericardial rub are non-specific post-op.

Causes

The causes of late cardiac tamponade are:

- Intrapericardial bleeding, e.g., following removal of pacing wires (prevented in some centers by leaving a small mediastinal drain in until after pacing wire removal), coagulopathy, or anticoagulation.
- Serous effusions due to hypervolemia, or post-cardiotomy pericarditis.

Diagnosis

Any of the physical signs listed in Box 6.2 may be present, together with an enlarged cardiac silhouette on the PA CXR. Confirmation is by urgent TTE or by TEE if views are inadequate. The location of the effusion should be described as well as the depth (anything >2cm is significant, but remember that large chronic effusions may not cause tamponade, while small acute ones may). Cardiac tamponade is present if there is RV diastolic collapse.

Management

- Secure iv access and give volume to restore BP.
- Place patient NPO, start 15min obs, and monitor with telemetry.
- The definitive management is urgent aspiration—pericardiocentesis for late simple effusions (>4 weeks) or subxiphoid pericardial window for bloody or loculated effusions (📖 p490). Within one or 2 days of surgery the management is urgent resternotomy (📖 p150).

Chest pain and ischemia

Chest pain

Chest pain is common after cardiac surgery, but it normally reflects the effects of a median sternotomy, rib retraction, and drains. Taking a careful pain history should help differentiate between the causes of chest discomfort listed in Box 6.3.

Box 6.3 Causes of postoperative chest pain

Dull, central ache

- Myocardial ischemia (usually brought on by exertion).
- Cardiac tamponade (📖 p143, p259).
- Pericarditis (classically soreness rather than ache).
- Peptic ulcer disease, esophagitis (worse on leaning forward and hot drinks, better with antacid), rarely pancreatitis.

Pain on movement

- Musculoskeletal pain, wound infection, unstable sternum.
- Chest drains.

Pleuritic pain

- Chest infection.
- Pneumothorax.
- Chest drain *in situ*.
- PE.

Diagnosis

Take a careful history and examine the patient. A CXR will demonstrate most lung pathology and a 12-lead EKG should help identify myocardial ischemia or pericarditis. Review previous medical history for peptic ulcer disease and the drug chart for NSAID use. Consider a TTE for persistent chest discomfort that is neither ischemic nor musculoskeletal in origin to exclude late cardiac tamponade presenting atypically, and a CT chest to identify mediastinitis if the patient has other evidence of infection.

Myocardial ischemia

Myocardial ischemia on ICU is covered on 📖 p185. Presentation later after cardiac surgery with myocardial ischemia may be due to factors listed in Box 6.4.

Box 6.4 Causes of post-op myocardial ischemia

- Incomplete revascularization.
- Early graft occlusion (usually related to technical issue).
- Vasospasm of arterial conduits (unusual once patient has left ICU).
- Trauma to native coronaries at or near annulus during valve surgery.
- Hypotension.
- Arrhythmias.
- Hypoxemia, profound anemia.

Diagnosis

Take a history: focus on dull pain related to exertion (ischemic) vs. sharp, positional, or pain worse on coughing. The physiotherapists may report bradycardia on exercising. A 12-lead EKG will confirm the presence of myocardial ischemia. Cardiac enzymes (CK-MB and troponin I and T) are often raised postoperatively, but serial measurements showing a continued or large rise suggest ongoing myocardial ischemia. The definitive investigation is cardiac catheterization to identify coronary obstruction.

Management

In patients with poor native vessels where incomplete revascularization was all that was possible, medical treatment is the mainstay. In patients with normal coronary arteries who underwent valve surgery, and in patients thought to have undergone complete revascularization, ongoing ischemia may need ICU management (📖 p185), otherwise:

- Ensure the patient is on aspirin 325mg od po.
- Ensure patients with radial artery grafts, or total arterial revascularization are given calcium channel blockers (diltiazem 60mg three times a day po) if BP will allow.
- Consider re-starting preoperative antianginal medication.
- Discuss indications for angiography with surgeons and a cardiologist.

Perioperative myocardial infarction

Box 6.5 contains the criteria for diagnosis of primary MI. Perioperative MI is difficult to diagnose because the patient may be unable to give a good history, or to distinguish between cardiac and non-cardiac chest pain; postoperative EKGs contain many non-specific changes, and heart block may mask ST changes; and because cardiac enzymes are usually raised postoperatively. New Q waves are diagnostic after the fact.

Box 6.5 Diagnostic criteria for myocardial infarction

In the setting of symptoms suggestive of acute coronary syndrome:

- EKG shows ST segment elevation: STEMI.
- No ST elevation, but elevated CK-MB (2 × normal) and troponin positive: non-Q wave or NSTEMI.

Causes

Graft occlusion (📖 p303) causes infarcts but up to 80% of perioperative infarcts are supplied by patent grafts. The causes of MI in these cases are incomplete revascularization, inadequate myocardial protection, emboli, and vasospasm.

Diagnosis and management

These signs suggest perioperative MI, which normally manifests on ICU:

- Hemodynamic compromise associated with:
  - ST depression, new Q waves (new T wave inversion is a common finding in postoperative EKGs and probably does not reflect MI) and a significant increase in CK-MB beyond usual postoperative levels.
- Regional reduction in wall motion on echo.

Management is described on 📖 p185.

Late arrhythmias

Atrial fibrillation

This protocol can also be used to treat multiple ventricular ectopics. Go and see the patient! If the patient is hemodynamically stable reassure them: AF happens to 1 in 3 patients after cardiac surgery. If patient is hemodynamically compromised (because of fast rate, hypertrophic LV following AVR for AS) organize DC cardioversion (📖 p224). Treat with:

- 10–20mmol KCl via central line to bring serum K^+ to 4.5–5.0mmol/L.
- 10–20mmol $MgSO_4$ via central line empirically.
- Restart β-blocker if this has been stopped, even if ↓BP.
- 300mg amiodarone over 1h via central line if mod to good LV.
- Followed by 900mg amiodarone over 23h iv, then oral dose.
- If poor LV give 250 micrograms digoxin iv over 30 minutes instead, and repeat digoxin to a maximum of 1500 micrograms, or until rate controlled.
- If no central access give potassium orally (e.g., 4 × tablets Sando-K) or peripherally (20mmol in 500mL N saline over 4h).
- Give magnesium sulfate orally.
- Load amiodarone orally (400mg three times a day for 24h).
- If the patient has longstanding AF and cardioversion is unlikely, digoxin may help gain rate control.
- Patients in AF should be on a LMWH and aspirin to prevent formation of mural thrombi, stroke, and PE.

Risk factors for atrial fibrillation postoperatively

- Age.
- Preoperative AF.
- Hypokalemia.
- β-blocker withdrawal.
- Inotropes especially dopamine, dobutamine, and isoprenaline.
- Digoxin toxicity can cause fast AF.
- Sepsis, especially chest infection.

Management of AF refractory to medical therapy

It is important to re-establish sinus rhythm in patients with new AF after cardiac surgery. Although AF is perceived as a largely benign arrhythmia, patients have a five fold increase in the incidence of neurological events postoperatively. 15% of patients in AF have been shown by TEE to develop mural thrombus within 48h (📖 p219). Long-term warfarinization is associated with a 0.5% pa incidence of intracranial hemorrhage (📖 p219).

- If a patient has been in constant AF for >36h but <72h despite optimal medical treatment they should undergo DC cardioversion under TEE guidance (📖 p224).
- If a patient has been in constant AF for over 72h, but has not been anticoagulated, they should be sent home on warfarin with a target INR of 2–2.5 and a reducing dose of amiodarone (200mg tid po 1 week, 200mg bid po 1 week, 200mg od po 4 weeks), reviewed in clinic

at 6 weeks and if they are still in AF they should be DC cardioverted, or if in sinus stop warfarin and amiodarone.
- If the patient is in paroxysmal AF a DC cardioversion is likely to have only temporary success: warfarinize and reassess in 6 weeks as described earlier.
- It is occasionally appropriate to attempt DC cardioversion for established AF in patients who have not been anticoagulated, providing that no mural thrombus can be identified on a TEE immediately prior to DC cardioversion.

Bradyarrhythmias

The diagnosis of bradycardias is explained on 📖 pp217 and 221. The commonest causes of bradycardias after cardiac surgery are:
- Drugs: β-blockers, calcium channel blockers, amiodarone, digoxin.
- Heart block, particularly post valve surgery.
- Patient's normal resting heart rate (e.g., fit, young patients).
- Hypoxia (late sign).

Management

Go and assess the patient: are they sitting up comfortably having tea, or are they unresponsive?
- If unresponsive follow the arrest protocol (inside front cover).
- Look for pacing wires and connect them to a functioning box (📖 p228).
- In an emergency give atropine 0.5 mg iv and repeat up to 3mg in total.
- In an emergency it is possible to pace externally (Fig. 5.18).
- If the patient is comfortable simply stop any iv amiodarone or digoxin, stop calcium channel blockers and β-blockers.
- Check electrolytes and blood sugar and treat accordingly.
- Get a 12-lead EKG to exclude new heart block or ischemia.
- Heart block postoperatively normally resolves within 3–4 days as hematoma and edema around the conducting pathways settle, and electrolyte abnormalities are stabilized, but if this does not happen discuss the patient with the cardiologists with a view to a pacemaker: they will usually request an event recorder or 24h tape.
- If a patient waiting for permanent pacemaker insertion needs anticoagulation use iv heparin.
- If a patient is transferred from ICU with no or minimal underlying rhythm, always make sure they are monitored with telemetry, and that you check the pacing thresholds daily (📖 p231). If these start to rise, the patient needs a permanent pacemaker or temporary transvenous pacing wires.

Indications for permanent pacemaker post-op

- Complete heart block.
- Mobitz type II heart block.
- Bradycardia <40bpm, associated with congestive cardiac failure, pauses >3s, or symptoms.
- Symptomatic bifascicular or trifascicular block.
- First-degree heart block in isolation is not an indication for a PPM.

Other cardiac problems

Hypertension

Once antihypertensive infusions such as NTG and propofol are discontinued, and myocardial function recovers from the insult of surgery, BP normally returns to preoperative levels. Preoperatively most patients have high BP controlled with a variety of oral medication including antianginals, most of which are discontinued in the perioperative period. Hypertension becomes a feature of most patients by day 3 postoperatively if appropriate drugs are not started. To treat hypertension restart preoperative antihypertensives at a lower dose, bearing in mind the following considerations:

- Patients should not need antianginals, e.g., nitroglycerin, isosorbide mononitrate, or nicorandil, and these are normally discontinued.
- ACE inhibitors (ramipril 2.5–10mg od po nocte) are started routinely in most patients for secondary prevention, and used in patients with poor LV to reduce afterload.
- ACE II inhibitors (losartan 40mg bid po) are used in patients where the dry ACE inhibitor cough is problematic.
- ACE inhibitors are not restarted in patients with hypotension or impaired renal function, but this may be reviewed when the patient is seen in outpatients.
- Start β-blockers, e.g., metoprolol 12.5–75mg po bid for secondary prevention, arrhythmia prophylaxis, and BP control.
- Patients with radial artery grafts are usually on a calcium channel blocker (diltiazem 60mg tid po for 6 weeks) to prevent vasospasm.
- In patients with tachyarrhythmias and good LV function, β-blockers are the first-line treatment for hypertension, but calcium channel blockers can also work well in this context.

Pericarditis

Pericarditis post-cardiotomy (also known as post-pericardiotomy and Dressler's syndrome) affects about 5% of patients. It is commoner in young patients. It may occur within a week of surgery or be delayed by several months.

Diagnosis

- The classical findings are constant chest pain that is sore rather than dull, pyrexia, and a pericardial rub.
- Malaise, lethargy, myalgia, and arthralgia often feature.
- WCC, ESR, and CRP are often raised.
- Pericardial and/or pleural effusions may be noted on CXR and echocardiography.
- ST segments are elevated concave upwards in all leads.

Treatment

- Give NSAIDs (ibuprofen 400mg tid po for 6 weeks).
- Prednisolone (10mg od po) has been used successfully in cases refractory to NSAIDs. Associated pericardial effusion should be treated as appropriate (📖 pp259, 265).

Pericardial effusion

Many patients have small pericardial effusions postoperatively which resolve without causing cardiac tamponade. In about 5% of patients these effusions increase in size. If the patient is not hemodynamically compromised by cardiac tamponade (📖 p259), these effusions may be treated by starting or increasing diuretic therapy.

- Most patients are already on furosemide 40mg po od, and this may be increased in increments up to 80mg po bid.
- If this fails to correct edema and effusions, consider adding spironolactone 25mg po od (a potassium-sparing diuretic which is particularly effective in the setting of ascites) or a short course of bumetanide 2mg od po for 72h or metolazone 5mg od po for 72h (these are thiazide diuretics which are effective at reducing refractory fluid overload but can impair renal function).
- Correct hypoalbuminemia by securing a dietician review to assess nutrition and prescribing protein food and drink supplements (cartons of Fortisips, Fortijuice, or Ensure with every meal) or NG feeding.
- Cardiac tamponade must be actively excluded and, if present, treated (📖 p259).

Constrictive pericarditis

Constrictive pericarditis complicates about 0.1% of cardiotomies, and is a late sequelae of up to half of cases of post-cardiotomy syndrome. A hemorrhagic pericarditis progresses over a matter of months to years to a thickened, fibrotic, and frequently calcified pericardium that tamponades the heart (📖 p488). Other risk factors include mediastinal irradiation and postoperative hemopericardium. The diagnosis and management is described on 📖 pp488–489.

Complications of valve surgery

See Table 6.4.

Table 6.4 Overview of the complications of valve surgery

Complication	Mechanical valve	Bioprosthetic valve
Infective endocarditis	0.5% within 30 days, 2% at 5 years.	0.2% within 30 days, 2% at 5 years
Prosthesis failure	Negligible	Aortic 30% at 15 years, mitral 60% at 15 years
Bleeding problems	2% of patients will have a major bleeding episode pa. Mortality 1.3% pa Thromboembolism: 2% pa	Bleeding events rare, thromboembolism 1% pa Annual stroke rate lower than for mechanical valves
Heart block	13% transient. 2% permanent Uncommon in MVR	13% transient. 2% permanent Uncommon in MVR
Survival	No difference	No difference

Infective endocarditis

Infective endocarditis complicates 0.2–0.5% of mechanical valve replacements within 30 days of surgery, 2% at 5 years, and 1% pa thereafter. The figures for tissue valves are similar, but slightly less at 30 days. Prosthetic endocarditis is described on 📖 p418.

- Blood cultures should be taken in any valve patient with a temperature >36.5°C more than 3 days postoperatively.
- Standard advice regarding antibiotic prophylaxis for invasive procedures should be adhered to from day 1 after surgery, not just following discharge: bladder catheterization should be covered by one dose of gentamicin 120mg im, for example.
- Any potential source of bacteremia should be treated aggressively: infective endocarditis can result from iv cannula cellulitis, superficial wound infections, urinary and respiratory tract infections, and most commonly infections related to arterial and central venous catheters.

Paraprosthetic leak

Incidental paraprosthetic leaks occur in about 2% of aortic valve replacements and 1% of mitral valve replacements. Paraprosthetic leaks within 30 days of surgery are usually either the result of the sewing ring being poorly seated within a calcified, irregular annulus; or the result of a suture cutting out of the annulus. Wash jets, the regurgitant jets typical of the valve mechanism, may be mistaken for paravalvular leaks.

- Late paravalvular leaks are usually associated with endocarditis, in which case they almost inevitably require surgical repair after treatment with antibiotics.
- High-velocity paravalvular leaks may cause hemolysis (📖 p289) severe enough to be an indication for valve replacement.
- If neither hemolysis or infective endocarditis is present, the management of the paravalvular leak depends on the degree of hemodynamic compromise: small leaks that do not progress can be treated conservatively for many years.

Prosthesis failure

Structural problems, not associated with thrombosis, endocarditis, hemolysis or hemorrhage, may necessitate valve replacement.

- Structural failure of mechanical valves is rare: the few valves that have had a higher than expected incidence of problems, such as the silver-coated sewing ring of the Silzone valve, have been withdrawn.
- Bioprostheses last longer in the aortic position and older patients than in the mitral position and younger patients (📖 p366): a 70-year-old having bioprosthetic AVR is very unlikely to experience bioprosthesis failure, whereas a 50-year-old has a roughly 50% lifetime risk of prosthesis failure.
- Patients are reviewed yearly, either by the cardiologists or by the surgeons, with a TTE to identify and monitor valve failure.
- The indications for surgery for structural failure of a bioprosthesis are similar to those for native valves (📖 pp376, 342, with a higher threshold for intervention according to the risk of reop surgery).

Bleeding and thromboembolism

Thromboembolic events include TIA and stroke. Valve thrombosis is obstruction of the mechanism of the prosthesis due to thrombus.

- The incidence of thromboembolism is the same in patients with tissue valves as it is in appropriately anticoagulated patients with mechanical valves: 1–2%% per patient per year. Patients with mechanical valves have a higher stroke risk because there is a 1–2% pa risk of major hemorrhagic event including strokes associated with anticoagulation.
- Valve thrombosis is most commonly due to inadequate anticoagulation because of poor compliance, changes in other medication and illness: 95% of cases of valve thrombosis therefore occur in mechanical valves.
- Smaller aortic valves may be at higher risk than larger valves.
- Patients with prosthetic valve thrombosis may present with angina, pulmonary edema, poor peripheral perfusion and systemic embolization, and acute hemodynamic compromise. TTE, TEE, fluoroscopy or CT may be needed to show failure of one or both leaflets to open.
- Thrombi <5mm that are not obstructing the valve orifice or mechanism may be treated with formal anticoagulation alone. Thrombolysis is rarely effective in chronic organized thrombus: the treatment is emergency reoperative valve replacement with a bioprosthesis.

Complete heart block

About 15% of patients undergoing isolated mitral or aortic valve replacement suffer transient heart block postoperatively, and so most surgeons routinely place temporary epicardial pacing wires.

- Heart block normally settles after 3–4 days as hematoma and edema resolves: persistence beyond this should be discussed with a cardiologist in case a PPM is indicated (📖 p263).
- About 1% of valve surgery patients will require permanent pacemaker insertion (📖 p263): this is a day case procedure and should not delay discharge, but may be postponed until the INR is <1.5.
- The preoperative risk factors for PPM are not well defined but probably include preoperative heart block and heavily calcified valves.

Respiratory complications

Chest infection

Incidence of pneumonia after cardiac surgery varies from 2 to 20%. The variation in incidence reflects variation in diagnostic criteria.

Diagnosis

A history of cough with purulent sputum, or purulent secretions aspirated from the ET tube, suggests chest infection. Pyrexia, bronchial breath sounds, reduced air entry, leukocyte neutrophila, raised CRP, and signs of consolidation on CXR confirm the diagnosis. Sputum culture may yield sensitivities of causative organisms. In the dyspneic, hypoxic patient perform arterial blood gases to guide immediate management.

Management

There is no good evidence that preoperative physiotherapy helps to prevent chest infection after surgery. The single most important intervention is to prevent patients with active chest infections undergoing surgery: any elective patient with a current cough (dry or productive), temperature, clinical signs of chest infection, neutrophilia, or suspicious CXR should be deferred for a fortnight and then reassessed. Other risk factors include active smokers, or patients who have stopped smoking within the last 6 weeks, patients with COPD, obesity, patients requiring prolonged ventilation, and patients who aspirate, particularly those recovering from stroke or with a vocal cord paresis.

- Physiotherapy helps patients expectorate, preventing mucus plugging.
- Effective analgesia is important to allow patients to cough.
- Culture sputum and blood, bronchoscope intubated patients.
- Empiric antibiotic cover varies between institutions but ciprofloxacin 250mg bid po provides good Gram –ve and +ve cover until organism sensitivities are known. Treat aspiration pneumonia with iv cefuroxime 1g tid and iv metronidazole 500mg tid.
- If the patient requires oxygen (PaO_2 <8.0kPa on room air) humidifying it reduces the risk of mucus plugs, and makes secretions easier to shift.
- The hypoxic, tachypneic, tiring patient on maximal respiratory support should be reviewed urgently by a senior for reintubation.

Pulmonary embolism

Postoperative PE is rare (<2% of cases) because bypass results in residual heparin effect, hemodilution, thrombocytopenia, and platelet dysfunction; and LMWH and TED stockings are used universally. The diagnosis and management is discussed in more detail on 📖 pp476–477.

Vocal cord problems

The incidence after cardiac surgery is probably about 1–2%. Patients present with hoarseness, breathlessness, difficulty phonating and expectorating, and coughing after drinking. These patients should be placed NPO to minimize risk of aspiration pneumonia, be reviewed by an ENT surgeon (vocal cord injection may be required) and undergo a speech and swallow assessment.

Exacerbation of COPD

The incidence of COPD in cardiac surgical patients is about 5%.

- Most studies show that mild to moderate COPD is not associated with a significant increase in postoperative complications, mortality, or length of stay.
- Severe COPD and preoperative steroid use has an associated incremental mortality of around 2% following cardiac surgery, with significantly poorer medium- and long-term outcomes.
- Ensure that all patients are routinely prescribed regular postoperative nebulizers (saline 5mL prn, salbutamol 2.5–5mg 4 times a day prn and becotide 500 micrograms 4 times a day prn) which also reduce risk of mucus plugging.
- β-blockers may safely be recommenced, but beware of starting β-blockers for the first time in patients with asthma or severe COPD.
- In hypoxic patients with COPD give maximal oxygen by CPAP if necessary and monitor blood gases: do *not* restrict oxygen for fear of reducing hypoxic respiratory drive, as in the acute setting hypoxia is a far more important complication than problems due to hypercapnia.

Pleural effusion

Up to 40% of patients undergoing cardiac surgery develop pleural effusions which may be unilateral or bilateral. IMA harvesting is associated with a higher rate of ipsilateral pleural effusions: up to 75% of patients are affected. It is unclear if pleurotomy at the time of IMA harvest, or taking pedicled rather skeletonized IMA, increase the risk of pleural effusion.

Diagnosis

Often asymptomatic. Dyspnea, pleuritic chest pain, decreased air entry, dullness to percussion suggest pleural effusion. It may be difficult to distinguish between consolidation and effusion on a CXR, but a meniscus is suggestive of pleural effusion and an air–fluid level diagnostic of a hydropneumothorax. If in doubt US or chest CT will quantify the size of the effusion and locate a safe area and direction for drainage.

Management

- Small pleural effusions can be treated with increasing diuretic therapy (furosemide 40–80mg bid).
- Large pleural effusions (>3–4cm deep on US) should be drained with a pigtail catheter (📖 p249) if late or a formal chest drain (📖 p248) if early and likely to be predominantly blood and clots.
- Leave the drain in on suction until <100mL drains in 24h.

Phrenic nerve paresis

Phrenic nerve paresis after cardiac surgery is common, although it may not manifest clinically. Although direct trauma particularly during IMA harvest, traction from the sternal retractor, SVC cannulation, and diabetic neuropathy have all been suggested as causes, the single most important factor is probably cold injury from slush placed in the pericardial cradle for topical cooling. Unilateral injury is of limited importance in patients with good respiratory function. Diagnosis and management of bilateral phrenic nerve palsy is discussed on 📖 pp748–749.

Renal complications

Acute renal failure

The etiology, diagnosis, and management of renal failure on ICU is described on 📖 pp232–238. Patients with elevated creatinine and urea who have returned to the ward should be weighed daily and have daily serum electrolytes monitored. If renal markers deteriorate:

- Insert a urinary catheter and commence hourly fluid balance chart.
- Recheck the drug chart for and stop nephrotoxic medication, e.g., ACE inhibitors, NSAIDs, and antibiotics such as teicoplanin and gentamycin.
- Actively look for and exclude relative hypotension, hypovolemia, sepsis, and delayed cardiac tamponade as reversible causes.
- If the patient is hypovolemic stop diuretic therapy and hydrate carefully. If fluid overloaded give 80mg iv furosemide.
- Ask yourself if the patient needs readmission to ICU for central venous monitoring and inotropic support.

Refractory peripheral edema

The pathophysiology underlying the state of fluid and salt overload in most patients is described on 📖 pp106–107. In most patients this corrects with a large diuresis, encouraged with loop diuretics, over the first few days postoperatively, but in a minority significant fluid overload persists. If patient weight remains >3kg above preoperative weight:

- Check serum electrolytes and treat appropriately (following sections).
- Check albumin and treat with daily protein supplements if low: this will take days to weeks to correct.
- If edema is refractory to furosemide increased to up to 80mg po bid, consider a stat dose of bumetanide (2mg po) or metolazone (5mg po) which may be repeated if creatinine is not elevated.
- Spironolactone 25–50mg po od is a potassium-sparing diuretic that is effective for management of ascites and pleural effusions.
- Consider fluid restriction of 1500mL per 24h.
- Treat symptomatically with TED stockings and elevation of legs.
- Unilateral peripheral edema may be caused by DVT, cellulitis, extravastion of iv fluids, stroke, hematoma, and saphenous vein harvest.

Hypokalemia

Hypokalemia (K^+ <4.0) is common and as it predisposes patients to arrhythmias, such as AF and digoxin toxicity it should be treated. Hypokalemia is normally caused by diuretic therapy, insulin sliding scales, diarrhea and vomiting, steroids and poor nutrition. Acute severe hypokalemia (K^+ <3.0) in postoperative cardiac patients may result in life-threatening arrhythmias. It can be recognized by small or inverted T waves, depressed ST segments, prolonged PR interval, and U waves on the EKG (Fig. 5.13d).

- Educate the patient about which foods are rich in potassium (bananas, prunes, apricots, tomatoes, orange juice) and ensure availability.
- Change furosemide to co-amilofruse 5/40 or 2.5/20 which contains furosemide (either 40mg or 20mg) and amiloride (5mg or 2.5mg) which is a potassium-sparing thiazide diuretic.

- Add oral potassium supplements up to 160mmol daily (1 tablet of Sando K^+ contains 20mmol of K^+, 1 tablet of Slow K^+ which is better tolerated by most patients contains 12.5mmol KCl).
- If a central line is in place give 20mmol KCl in 50–100mL of 5% dextrose over 20min to 1h. Never give a bolus of KCl.
- If it is necessary to use a peripheral line place a maximum of 40mmol of potassium in 1L 5% dextrose running at a maximum of 125mL/h.
- Monitor KCl daily, and avoid discharging the patient home on a combination of potassium supplements and potassium-sparing diuretics as rebound hyperkalemia will be a risk.

Hyperkalemia

Hyperkalemia (K^+ >5.0) is seen in the setting of renal failure, tissue necrosis, and potassium-sparing diuretics and supplements. Acute hyperkalemia (K^+ >6.0) can cause life-threatening ventricular arrhythmias. EKG changes that herald myocardial dysfunction are flattened P waves, wide QRS complexes, tenting of T waves, and in peri-arrest hyperkalemia a sine wave appearance (Fig. 5.13c).

- Treat the patient with EKG changes as an emergency.
- Give 50mL of 50% dextrose containing 15U of Actrapid insulin as an infusion over 10–20min, repeating as necessary, checking blood sugar and K^+ after each infusion (use blood gas machine).
- Give 10mL calcium gluconate 10% iv over 2min, repeat.
- Calcium Resonium or Kayexalate® enema binds K^+ and removes it from the body.
- Dialysis should be urgently considered in patients with refractory hyperkalemia despite these measures, irrespective of renal function.

Hyponatremia

Hyponatremia (Na^+ <135mmol/L) occurs in fluid overload where patients have been given 5% dextrose fluid maintenance over long periods rather than normal saline, as a result of diuretic therapy, diarrhea and vomiting, renal failure, cardiac and liver failure, and SIADH (syndrome of inappropriate ADH secretion). Profound hyponatremia will lead to confusion and fits as a result of potentially life-threatening cerebral edema. Assess the patient for dehydration or edema, and measure urine osmolality and urinary sodium. SIADH is diagnosed by finding concentrated urine (sodium >20mmol/L) in the setting of hyponatremia; or hypo-osmolar plasma (<260mmol/kg), and low serum cortisol without hypovolemia.

- If the patient is edematous restrict daily fluid intake initially to 1.5–1.0L.
- Stop thiazide diuretics, but continue loop diuretics.
- If dehydrated give 1L normal saline iv over 8–12h.
- SIADH occurs in 1–2% of cardiac surgery patients as a result of increased secretion of vasopressin (or ADH) and should be treated by a combination of fluid restriction for several days. Loop diuretics, and sodium supplements may help.

Disturbances of acid–base balance

These are discussed on 📖 pp168–171.

Gastrointestinal symptoms

Abdominal pain

Distinguishing between peritonitic and colicky pain is particularly difficult in elderly, sedated, or confused patients following cardiac surgery. Classic signs of peritonitis, and localizing signs are frequently not present despite major intra-abdominal pathology. Take a full history, note recent drug regimes, examine the patient, and request CBC, biochemistry, LFTs, amylase, CRP, blood gases, abdominal plain film, and erect CXR in the first instance. Stool and blood cultures should be taken.

Differential diagnosis of post-op abdominal pain

- Constipation or paralytic ileus.
- Wound pain, muscular pain from coughing.
- Myocardial ischemia, lower lobe pneumonia.
- Peptic ulceration.
- Pancreatitis.
- Gallbladder pathology including cholecystitis, biliary colic.
- Small and large bowel obstruction.
- Ischemic bowel.
- Perforated viscus (chest tubes, TEE, ischemia).

Constipation

Failure to pass stool by day 4 postoperatively is not uncommon. Constipation may be caused by reluctance to use a bed pan, lack of privacy, immobility, pain on straining from wounds or anal fissures, dehydration, poor nutrition, inadequate dietary fiber, opiate analgesia, iron supplements, tricyclic antidepressants, and spinal anesthesia. Treatments are:

- Bulking agents, such as increasing dietary fiber with bran or dried fruit, methylcellulose 500mg tid, all of which increase stool bulk and decrease transit time.
- Stool softeners such as liquid paraffin, glycerin suppositories 1–2 od pr and phosphate enemas 1–2 od pr.
- Osmotic agents such as lactulose 10–20mL bid or magnesium hydroxide 50mL per 24h po, which should be taken with plenty of water.
- Stimulants such as senna 1 tablet bid po, bisacodyl 5–20mg nocte po, sodium docusate 60mg od po, co-danthramer 50mg po od, which should not be used in the long term as tolerance develops.

Diarrhea

Liquid stools may be caused by preoperative conditions such as inflammatory bowel disease or irritable bowel, or acute GI pathology including ischemic bowel and GI bleeding, but is more commonly due to inappropriate prescribing of laxatives, antibiotic therapy, digoxin, and cimetidine.

- Pseudomembranous colitis is caused by overgrowth of *Clostridium difficile* following antibiotic therapy, particularly with cephalosporins: send stools for culture and treat positive *C. diff* toxin diarrhea with either oral metronidazole (400mg tid) or oral vancomycin (125mg qid).

Nausea and vomiting

This affects up to 75% of patients. It predisposes to increased bleeding, incisional hernias, aspiration pneumonia, reduced absorption of oral medication, impaired nutrition, and metabolic abnormalities such as hypokalemia. Causes include:

- Bypass, prolonged surgery, anesthetic agents, e.g., etomidate, ketamine, N_2O, intraoperative and postoperative opioids, spinal anesthesia, gastric dilatation from ineffective bag and mask ventilation CPAP.
- Postoperative ileus, bowel obstruction, constipation, gastric reflux, peptic ulceration or bleeding, medications including many antibiotics, NSAIDs, opiates, statins, pancreatitis, sepsis, and hyponatremia.

Treat by withholding opiate analgesia (use regular paracetamol and NSAIDs if no contraindications), starting regular antiemetic, e.g., ondansetron, adding a second with different mechanism of action if needed, mobilizing patient. Treat ileus, monitor and correct fluid and electrolyte losses.

Classification of antiemetics

- Combination of two different types of antiemetic increases efficiency.

Antidopaminergic agents

- Active against emetic effect of opiods. Sedative. Extrapyramidal side effects—avoid in Parkinson's disease.
- E.g., prochlorperazine 12.5mg im, droperidol 1–1.25 mg iv, metaclopramide 10mg iv/im/po tid.

Antihistamines

- Sedation, tachycardias, hypotension with iv injection.
- E.g., cyclizine 50mg im/iv/po tid.

Anticholinergics

- Active against emetic effect opioids. Sedation, confusion, dry mouth
- E.g., hyoscine (scopolamine) 0.3–0.6mg im.

Antiserotonergics

- Lowest side effect profile of all antiemetics. Effective in chemotherapy nausea. Quite useful for postoperative nausea and vomiting.
- E.g., ondansetron 1–8mg po/iv/im tid, granisetron 1mg po/iv tid.

Paralytic ileus

Paralytic ileus occurs in up to 5% of patients after cardiac surgery. It may share features with small bowel obstruction, including nausea, vomiting, constipation, absent bowel sounds, and dilated loops of gas-filled bowel on abdominal plain film: usually patients are pain free, and passing flatus.

- Possible causes include gastric distension from ineffective bag and mask ventilation or CPAP, splanchnic and mesenteric ischemia, venous congestion, pseudomembranous colitis, drugs particularly opiates, other GI pathology including peptic ulceration, cholecystitis, and pancreatitis.
- Pass an NG tube to decompress the stomach and consider a flatus tube.
- Stop opiates and contributing drugs, and mobilize patients if possible.
- Put the patient NPO and give normal saline 80–125mL/h iv.
- Prolonged ileus and pseudo-obstruction may necessitate colonoscopy.

Gastrointestinal complications

Serious gastrointestinal complications occur in about 2% of patients following cardiac surgery, with previous history of GI pathology, age, severe atherosclerosis, or vascular disease all risk factors. The diagnosis is more difficult in postoperative, sedated, confused, and elderly patients, but the management is much the same as in non-surgical patients. Half of these patients undergo laparotomy: the mortality is up to 20%.

Upper gastrointestinal bleeding

Upper GI bleeding is most commonly caused by peptic ulcer disease, followed by gastritis, esophageal varices, esophageal tears and occasionally trauma to the esophagus or oropharynx from intubation with NG tubes, and TEE probes. Peptic ulceration risk after cardiac surgery is increased by the stress of major surgery, anticoagulation, NSAIDs, coagulopathy, and splanchnic hypoperfusion.

Diagnosis

- Hematemesis (may be via NG tube) more commonly occult.
- Melena, or positive fecal occult blood (FOB) test (guaiac).
- Abdominal pain, peritonism, hypotension, tachycardia.
- Drop in Hb (1g/dL per unit of blood lost) may be only evidence.

Prophylaxis

- Prophylaxis with omeprazole 20mg po od or omeprazole 40mg iv infusion over 2h should be given to patients with a history of peptic ulcer disease, gastritis, or hiatus hernia.
- Start NG feeding patients that fail to extubate early: 10mL h NG feed offers gastric protection reducing translocation of endotoxin.
- Avoid NSAIDs in patients with any history of dyspepsia or elderly.

Management

- Secure iv access with two large gauge (grey or brown) cannulas in the antecubital fossa, or use a Swann sheath if present for rapid volume replacement (a triple lumen is insufficient as it is long and narrow).
- XM 4 units of blood
- Transfuse blood to bring the Hb back to 8–9.0g/dL, titrating the rate of transfusion against BP, pulse, and CVP if available.
- Check coagulation and correct any coagulopathy.
- Discuss with gastroenterology with a view to esophagogastroduodenoscopy (EGD) ± injection of bleeding points.
- In patients on warfarin for mechanical valves restart heparin (APTT 50–70) once active bleeding subsides or has been treated.

Lower gastrointestinal bleeding

Lower GI bleeding is characterized by fresh rectal blood, but may require passage of an NG tube or even EGD to exclude an upper GI source. Common causes are mesenteric ischemia, infective colitis including *C. difficile*, angiodysplasia (Heydes' syndrome when associated with aortic stenosis), and bleeding from colonic polyps. Mesenteric angiography may be necessary to identify the lesion before surgical resection.

Mesenteric ischemia

Mesenteric ischemia may be caused by global hypoperfusion due to low cardiac output states, high doses of pressors, atherosclerotic embolism, thromboembolism, and associated with sepsis, DIC, or HITT. It presents on the ICU as worsening metabolic acidosis, failure to absorb NG feeds, and abdominal distension. Awake patients may complain of severe abdominal pain, nausea, and vomiting. Diarrhea which may initially be bloody is often present. Bowel sounds are absent. Peritonism is a late sign suggesting perforation. Diagnosis on clinical grounds is difficult in the paralyzed and ventilated patient. Contrast CT or angiography is useful. Definitive management is laparotomy and resection of infarcted segments with or without primary anastomoses. Not infrequently infarcted areas of bowel are so widespread that surgery is 'open and close' only.

Bowel obstruction

Small and large bowel obstruction post cardiac surgery is most commonly due to adhesions from previous surgery, followed by incarceration of hernias (inguinal, femoral, epigastric, incisional). Rare causes include occult carcinoma and pseudo-obstruction. Small bowel obstruction, which requires urgent general surgical review and surgery if not resolved within 24–48h, is characterized by colicky abdominal pain, nausea, vomiting, and distended loops of small bowel on abdominal X-ray (central valvulae conviventes cross the entire lumen of bowel which measure >6cm in diameter). Large bowel obstruction may usually be managed conservatively for longer.

- Place the patient nil by mouth, pass a NG tube.
- Start iv fluids (normal saline 1L with 40mmol K^+ over 8–12h.
- Discuss with general surgery.
- Reassess for signs of peritonism every few hours, check all labs daily.

Pancreatitis

A transient small rise in serum amylase is seen in over 1/3 of patients, pancreatitis occurs in 1%. It is caused by splanchnic hypoperfusion, but systemic emboli, prolonged use of vasopressors may contribute and activation of inflammatory pathways on bypass also contribute. Preoperative actors such as steroid use, gallstones, alcohol abuse, and hyperlipidemia are less important. Any patient with abdominal pain should have a serum amylase check. Complications of pancreatitis include shock, ARDS, renal failure, $\downarrow Ca^{2+}$, and DIC. Management is conservative:

- Give colloid to replace third-space fluid, with hourly fluid balance.
- Give morphine.
- Inotropic support is required in severe cases.
- Discuss with general surgeons.

Cholecystitis

This has an incidence of up to 1:200 after cardiac surgery, with patients presenting up to 2 weeks after surgery with fever, nausea, abdominal pain. Diagnosis is by US, and treatment is medical (iv fluids, nil by mouth, and antibiotics). The etiology is ischemic injury and cholecystectomy is often indicated.

Liver failure

Hepatic failure

Approximately 10% of patients develop mild hepatic dysfunction after cardiac surgery with a transient rise in liver enzymes and bilirubin. Fulminant hepatic failure, which complicates <0.1% of cardiac surgery patients with no known preoperative hepatic dysfunction, has a mortality of about 80%.

Etiology

- *Preoperative dysfunction*: hepatitis, cirrhosis, biliary obstruction.
- *Hemolysis*: blood transfusion, haematoma, paravalvular leak, glucose 6-phosphate dehydrogenase deficiency, prolonged bypass, excessive use of pump suckers.
- *Hepatocellular injury*: exacerbation of preoperative disease, low cardiac output states, splanchnic ischemia, prolonged high-dose pressors (causes splanchnic hypoperfusion), right heart failure, hepatotoxic drugs (acetaminophen, halothane, antiretrovirals).
- *Cholestasis*: intrahepatic (hepatitis, hepatocellular necrosis, cholestasis, parenteral nutrition, sepsis, drugs including opiates), and extrahepatic (biliary tract obstruction).

Diagnosis

Fulminant hepatic failure is acute hepatic failure occurring within 8 weeks of exposure to the insult, and complicated by hepatic encephalopathy. In addition to increases (in the high thousands) in AST, ALP, LDH and several-fold increase in σ-GT and bilirubin. Features of hepatic failure include:

- Jaundice: secondary to conjugated hyperbilirubinemia.
- Hepatic encephalopathy.
- Coagulopathy: secondary to ↑PT, ↓platelets and platelet function, DIC.
- Ascites.
- Sepsis, including increased susceptibility to fungal infections.
- Acid–base disturbance: commonly metabolic acidosis due to ↑lactate.
- Metabolic derangements: commonly ↓Na.
- Hypoglycemia due to glycogen depletion.
- Cerebral edema and raised intracranial pressure.
- Hepatorenal and hepatopulmonary failure.
- Reduced systemic vascular resistance.
- Arrhythmias including VF.

Management

Modest elevations in liver enzymes and bilirubin are usually transient and benign: no action other than reducing hepatotoxic medication is usually required. Management of fulminant hepatic failure is mostly supportive:

- Check blood glucose frequently and replace as indicated.
- Renal failure requires dialysis.
- Cardiovascular support requires PA catheter insertion and inotropes.
- Give empirical antibiotic prophylaxis.

Hepatorenal failure

Hepatocellular failure is commonly complicated by renal failure. This is attributed to:

- ↑peripheral vasodilation.
- ↑renal vasoconstriction.
- Resulting in ↓GFR and oliguria.

If there is no clear primary cause of renal failure, this is called hepatorenal syndrome. Evidence from transplantation suggests that the renal failure is entirely secondary to liver failure as the renal failure improves both after liver transplantation, and after transplantation of the affected kidney into an individual without liver dysfunction. The renal dysfunction is proportional to the severity of hepatocellular failure, and acute tubular necrosis may supervene. Therapy is supportive: careful management of filling and systemic vascular resistance. Dialysis may be required for refractory acidosis, hyperkalemia, and fluid overload.

Hepatopulmonary syndrome

Hepatocellular failure is less commonly complicated by hypoxemia, as a result of abnormal ventilation–perfusion ratios and diffusion capacity. This is attributable to:

- ↑ intrapulmonary vasodilatation.
- ↑PVR.
- Failure of oxygen to diffuse as readily into the centre of dilated vessels.
- ↑CO limiting the time for gas exchange.

Stroke

The incidence of stroke in cardiac surgery is about 1–2% for CABG, 2–3% for valve. Patients with one or more risk factors have a much higher incidence of stroke: this affects consent and operative planning.

Risk factors for stroke

- Increasing age (>80 years risk of CVA 3–5%).
- Diabetes.
- Previous history of stroke or TIA (increases risk three-fold).
- Carotid artery atherosclerosis.
- Perioperative hypotension.
- Calcified or very atherosclerotic ascending aorta.
- Calcified aortic or mitral valve.
- Left-sided mural thrombus.
- Cardiotomy.
- Long duration of CPB, DHCA.
- Postoperative AF.
- Failure to give antiplatelet therapy postoperatively.

Etiology

Embolic

- Microemboli, debris from operative field.
- Mural thrombus.
- Debris from valve excision, particularly calcified aortic valve.
- Septic emboli from endocarditis.
- Trauma to aorta from cannulation and clamping.
- Air embolism.
- Carotid atheroma.

Hemorrhagic

- Heparinization on bypass.
- Postoperative warfarinization.

Cerebral hypoperfusion

- Carotid and vertebral artery stenosis, dissection.
- Hypotension.
- Circulatory arrest (long period or insufficient cooling).
- Raised intracranial pressure.

Hypoxia

- Profound hypoxia before, during or after bypass for >3min.

Clinical features

Most neurological deficits are apparent once sedation has been weaned (i.e., within 24h), but 1/3 of strokes occur several days postoperatively. Any deficit resolving within 24h is called a transient ischemic attack (TIA). Clinical features include:

- Failure to regain consciousness once sedation has been weaned.
- Hemiplegia (middle cerebral artery or total carotid artery occlusion).

- Initial areflexia becoming hypereflexia and rigidity after a few days.
- Aphasia, dysarthria, ataxia, inadequate gag reflex, aspiration.
- Visual deficits, unilateral neglect, confusion.
- Persistent, marked hypertension, hypercapnia.

Diagnosis

The aim is to establish a definitive diagnosis, establish a cause to guide appropriate secondary prevention, and establish a baseline of function to help plan long-term rehabilitation, or withdrawal of therapy.

- Carry out a full neurological examination (cognitive function, cranial nerves, and tone, power, reflexes, and sensation in all four limbs).
- Modern contrast head CT will show infarcts within 2h (older scanners may not pick up lesions until they are 2–3 days old). You must distinguish between hemorrhagic and ischemic CVAs (1 in 10 are hemorrhagic): MRI is necessary to image brainstem lesions.

Prevention

Carotid endarterectomy

The single most important preoperative intervention is identifying significant carotid artery disease. All patients with a previous history of stroke or TIA, or with a carotid bruit should undergo Doppler ultrasonography of their carotids. Some surgeons Doppler all patients over the age of 65, and all patients with other severe extracardiac arteriopathy.

- Patients with a stenosis of their internal carotid artery <75% have a greater risk of stroke from carotid endarterectomy than they have from cardiac surgery alone, and their carotid disease should be left.
- Patients with a stenosis of >75% but <99% of their internal carotid artery and a history of TIA or stroke have a lower risk of stroke if they undergo carotid endarterectomy (or stenting in suitable subjects).
- Generally the overall risk of death or stroke seems lowest if carotids are treated after cardiac surgery, rather than before or as a combined procedure, which also avoids the problem of needing to stop the antiplatelet medication required after carotid stenting.

Operative technique

- Maintain MAPs 80mmHg in patients at risk of ischemic strokes.
- Selection of cannulation sites can be guided by intraoperative US of the ascending aorta, which identifies diseased plaques at risk of embolization, and is much more accurate than digital palpation.
- Meticulous removal of debris from the operative field in open heart (especially aortic valve surgery) surgery, exclusion or air from all arterial lines, and thorough de-airing is a key part of stroke prevention.
- 'No touch' technique in coronary artery bypass grafting (📖 p321).
- Systemic cooling provides some additional cerebral protection.
- Limiting use of pump suction reduces microemboli.

Postoperative measures

- Aspirin (and clopidogrel after CEA) and BP control.
- Again, maintain MAPs >80mmHg.

Management of stroke

Initial management

- Assess the airway, breathing, and circulation (📖 inside front cover).
- If the patient is unable to maintain their airway insert a Gudel airway, bag and mask ventilate with 100% O_2, and call anesthesia.
- Monitor BP, but do not attempt to correct high pressures as these are critical for adequate cerebral perfusion.
- Monitor oxygen saturations.
- Secure iv access and give colloid if indicated.
- If the patient is able to maintain their own airway and is not hemodynamically compromised, explain what has happened and reassure.
- Perform a full neurological examination.
- Put the patient NPO if there is no gag reflex.
- Send complete blood count and chemistry, and coagulation.
- Send blood cultures if there is any history of endocarditis, pyrexia.
- Get an EKG (identify AF, heart block, or other causative problem).
- Request an urgent CT head to identify emboli/hemorrhage.
- Unless there is a strong suspicion that the stroke is hemorrhagic, aspirin should be continued, as it has a negligible effect on the outcome of hemorrhagic stroke, but a small positive effect on the outcome of ischemic stroke.

Hemorrhagic stroke

Hemorrhagic stroke (including subarachnoid and subdural hemorrhage and intracranial bleeds) should be suspected if there is a history of head injury, or major coagulopathy. CT confirmation of diagnosis should be obtained urgently as early neurosurgical management may be indicated. If the patient is being warfarinized for a mechanical valve prosthesis the risk of causing an embolic event by reversing anticoagulation (with vit K, FFP, or clotting factors) is <0.02% for each day that the patient is not anticoagulated. The risk of rebleeding is greatest within the first 24h. Anticoagulation should be stopped, and coagulopathy corrected. Restart anticoagulation within 1–2 weeks if the patient is stable.

Embolic stroke

Aim for MAPs 80–100mmHg to improve perfusion. Request TTE to identify source.

Longer-term management

Medical treatment

Cardiothoracic patients are not candidates for thrombolysis: recent major surgery is one of the exclusion criteria. Aspirin reduces death and dependency by 13 patients per every 1000 treated. There is no evidence that LMWH, unfractionated heparin, or warfarin improves outcome.

Secondary prevention

Aspirin, ticlopidine, dipyridamole, and clopidogrel reduce recurrences (relative risk reduction of about 13%). Warfarin and heparin reduce recurrence embolic strokes in AF.

Ventilation

Patients with severe strokes, particularly those involving the basilar artery territories, may need intubation and ventilation for respiratory failure resulting from several causes:

- Abnormal patterns of respiration including apnea, Cheyne–Stokes.
- Loss of protective pharyngeal reflexes leading to aspiration.
- Failure to clear secretions.

Weaning ventilation, including criteria for weaning and extubating, are discussed on 📖 pp164–165. There are three general approaches to weaning:

- Spontaneous breathing trials (SBT).
- Pressure support ventilation (PSV).
- Synchronized intermittent mandatory ventilation (SIMV).

Spontaneous breathing trials can be performed with the patient either on or off the ventilator, and usually involve a T-piece and humidified supplemental oxygen. SBTs last from 30min to 2h. Positive pressure support or CPAP is usually added. In SIMV the mandatory ventilation rate is reduced so that the patient has to increase spontaneous breaths. Tracheostomy (📖 p166) is useful in selected patients to improve comfort but does not increase the speed of weaning from ventilation, which can take weeks to months.

Intubation for apnea complicating stroke is a poor prognostic sign; most deteriorate neurologically, and few of these patients are successfully weaned from ventilation. Intubation primarily for airway protection has a more successful outcome. The majority of patients can be weaned from ventilation, but aspiration pneumonia may complicate recovery.

Feeding

Protective pharyngeal reflexes are impaired in stroke. The decision to remove a tracheal tube and to commence oral feeding is based on standardized swallow assessments, usually carries out by a speech and language therapist (SALT). NG feeding is mandatory until these tests are passed. Feeding is then graduated, starting with thickened fluids and progressing to solids.

- Swallow assessment consists of giving a decannulated patient who is alert and able to sit upright, consecutive teaspoons of water followed by a glass of water: the patient fails if at any stage they fail to actively swallow, or if it results in coughing, spluttering, or gurgly voice.
- The dye test consists of placing a small amount of blue dye in the mouth of the patient with an ET or tracheostomy tube *in situ*: the patient fails if dye is then found in bronchial secretions on suction.
- Videofluoroscopy (modified barium swallow) is the gold standard for diagnosis of continuing aspiration.
- A PEG (percutaneous endoscopic gastrostomy) tube is usually required for long-term feeding in a nursing home environment.

Mobilizing

Structured neurophysiotherapy and intensive nursing is vitally important in preventing complications such as pressure sores, strictures, constipation, and aspiration pneumonia as well as maximizing early return of function.

Neurological complications

Confusion

Confusion is common post-operatively. It is often obvious, with a disoriented, uncooperative, or hallucinating patient. Frequently, confusion is not overt, consisting of inactivity, quietness, slowed thinking, and labile mood, and it is only spotted by relatives or nursing staff. Actively assess whether the patient is oriented in time, person, and place. Perform a quick mini mental state examination if you are still unsure.

Management

- If the patient's behaviour poses a physical danger to himself or others, it may be necessary to sedate as first-line management: haloperidol 2.5mg may be given up to a total of 10mg in 24h po, im, or iv, but if the patient remains disturbed 2.5–5mg of midazolam should be given iv, and the patient placed under close observation.
- Beware of sedating the hypoxic or hypotensive patient as this may trigger a cardiorespiratory arrest.
- Treat respiratory failure (📖 p268) and low BP (📖 p258).
- Reassess the drug chart: stop opiates and benzodiazepines.
- Correct metabolic abnormalities.
- Alcohol withdrawal is diagnosed from a history of chronically high alcohol consumption often with raised σ-GT, combined with psychomotor agitation postoperatively. It can be treated with diazepam 5–10mg tid po/pr: haloperidol lowers the seizure threshold so is not first line therapy for confusion in these patients.
- Perform a neurological examination to look for focal neurological deficit and consider head CT to exclude stroke.
- Reassure patient and relatives: confusion is common, almost always reversible, and it is not a sign that the patient is 'going mad'.

Common causes of confusion

- Sleep deprivation, ICU psychosis.
- Medication (particularly benzodiazepines, opiates, anticonvulsants).
- Stroke, 'post-pump' (micro air emboli).
- Hypoxia, hypercapnia.
- Shock.
- Sepsis.
- Alcohol withdrawal.
- Metabolic disturbances (↓glucose, Na^+, pH, ↑Ca^{2+}, Cr, urea, bilirubin).
- Post-ictal.
- Pre-operative dementia.

Seizures

Seizures may be generalized (loss of consciousness with or without tonic-clonic features, absence seizures, atonia) or partial (localized to part of one hemisphere). Causes of seizures include:

- Physical: stroke, intracranial bleed, cerebral air emboli, septic emboli

- Metabolic: withdrawal of alcohol, anticonvulsants or benzodiazepines, (↓↑glucose, Na^+, pH, Ca^{2+}, ↑Cr, urea, bilirubin), tricyclic antidepressants, lidocaine toxicity.

Management

Status epilepticus is any seizure lasting >30min, or repeated seizures without intervening consciousness. Treat by securing the airway, give oxygen, suction if required, diazepam 10mg iv (or PR) over 2min, and then 5mg/min until seizures stop or respiratory depression occurs, then start phenytoin 15mg/kg iv up to 50mg per min if seizures continue. Most seizures are self-limiting: start regular phenytoin as prophylaxis and look for an underlying cause.

Peripheral nerve injuries

Brachial plexus injury

This is caused by indirectly by sternal retraction, and directly by trauma when placing central lines. It presents as paresthesia and weakness in the C8–T1, and if severe C6–C7. Formal diagnosis is by EMG studies. It can be avoided by placing the sternal retractor as caudally as possible and opening it slowly. Treatment is physiotherapy, with referral to a pain service for chronic pain unresponsive to analgesia.

Horner's syndrome

Ptosis, meiosis, anyhydrosis, and enophthalmosis resulting from damage to the sympathetic nerve supply to the eye at any stage from its journey from the central nuclei and spinal cord to post-ganglionic fibres via the stellate ganglion. Inadvertent damage during central line placement, carotid endarterectomy, harvest of the very proximal IMA, and stroke may cause this syndrome. Treatment is symptomatic.

Recurrent laryngeal nerve palsy

This may occur as a result of IMA harvesting, trauma to the arch of the aorta, e.g., from cannulating, trauma to the nerve in the neck from internal jugular vein cannulation, pressure injury from a malpositioned ET cuff, and cold injury. It presents as hoarseness and breathlessness after extubation. Patients should be seen by an ENT surgeon (📖 p268).

Phrenic nerve palsy

This is discussed on 📖 pp269, 748–749.

Sympathetic dystrophy (radial and saphenous nerve)

Injury to these nerves when harvesting conduit usually results in paresthesia (over the anatomical snuff box in radial nerve injury and over the medial malleolus and ankle in saphenous nerve injury). Occasionally this is associated with increasing pain, swelling, and trophic changes (hair loss, shiny skin). Sympathetic dystrophy is difficult to treat: it may respond to NSAIDs such as ibuprofen, regular analgesia, or amitriptyline for intractable neuropathic pain.

Ulnar nerve injury

Failure to protect the arms with padding intraoperatively may lead to ulnar nerve paresis from pressure injury to the ulnar nerve as it passes round the medial epicondyle of the elbow.

Wound infections

Deep sternal wound infection

The incidence of deep sternal wound infection is reported at 0.5–5%. As up to 75% of wound infections present after discharge from hospital, the incidence is frequently under-reported. Risk factors include:

- *Preoperative*: age, diabetes, obesity, smoking, steroid therapy, COPD.
- *Operative*: paramedian sternotomy, bilateral pedicled mammary artery harvest, prolonged surgery, poor surgical technique (including excessive bone wax and cautery to the periosteum, inaccurate placement of sternal wires, poor aseptic technique).
- *Postoperative:* re-sternotomy for bleeding, multiple transfusions, mediastinal bleeding, prolonged ventilation, and impaired nutrition.

Deep sternal wound infection (DSWI) definition

The Centers for Disease Control and Prevention (CDC) define DSWI as infection in tissue deep to the subcutaneous layers *and* either positive cultures, macroscopic/histological evidence, symptomatology, *and* pus.

Diagnosis

- Fever.
- ↑WCC, positive blood cultures.
- Sternal 'click' (ask the patient to cough, or turn their head left and right while feeling for movement in the sternum).
- Serous or purulent discharge, wound dehiscence.
- Pericardial effusion.
- CXR may show fractured or migrating sternal wires.
- CT may show evidence of non-union, pus, and osteomyelitis.

Prevention

- *Preoperative*: weight reduction in obese patients, reducing steroid therapy, optimizing respiratory function, screening for and treating MRSA. Shaving patients in the holding area and not the night before results in lower numbers of skin organisms.
- *Operative:* meticulous aseptic and operative technique. Separate leg and sternal wound instruments (sharing increases the incidence of *E. coli* sternal wound infection).
- *Postoperative*: appropriate antibiotic prophylaxis. One dose must be given on induction to ensure peak blood levels by the time of incision. Wash hands between examining every patient (alcohol rinse has been shown to be more effective than soap and water at reducing cross-infection).

Management

- A cough lock may reduce pain and reduce the impact of dehiscence.
- Iv antibiotics (teicoplanin 400mg od + tazocin 4.5g tid).
- Vacuum dressing.
- Surgical debridement and rewiring may be necessary.

Rewiring for sternal wound infection

This is never undertaken lightly. Patients may have multiple organ dysfunction as a result of sepsis, tissues are friable, and surgery often takes place at a time when adhesions are maximal (4–6 weeks postoperatively). The principles are the same as for an elective resternotomy (📖 p346). Ensure that the patient has appropriate antibiotics on induction, a radial arterial and central line. Devitalized tissue is debrided back to bleeding margins, including the sternal edges which need to be undermined sufficiently to place sternal wires. All pus is evacuated from the pericardium, paying great care to anastomotic lines, any of which may be involved in infection. Send pus and bone for culture to guide treatment and detect osteomyelitis. Often primary closure cannot be achieved due to extensive debridement: options include pectoralis major advancement flaps, transrectus abdominus muscle (TRAM) flaps, and rarely omental flaps. Plastic surgeons should be involved for these. There are many alternative closure techniques:

- Standard closure: residual osteomyelitis may flare into repeat infection.
- Betadine washout: adjunct to all techniques, of no proven benefit.
- Robicek closure, sternal bands or plates to improve stability.
- Irrigation systems: used in refractory cases, or for prosthetic material.
- Sternectomy and pec major or omental flap, most commonly staged with vacuum dressings: eradicates most persistent infections.

Superficial wound infection

The CDC defines SWI as infection involving only skin and subcutaneous tissue *and* either purulent drainage, positive cultures, or symptoms. The incidence of superficial sternal wound infection is about 3–5%, open saphenous vein harvest site infection up to 25%, and superficial radial wound harvest site infection about 2%. Diagnosis is usually based on local cellulitis or discharge. Swab the wound, take blood cultures if pyrexial. Management includes:

- Antibiotics, e.g., ciprofloxacin 500mg bid po for 1 week.
- Incision and drainage of any fluctuant areas.
- Vacuum dressing for deep wounds with large amounts of pus.
- Monitoring for any signs of deep sternal wound infection.

Brief guide to antibiotic cover

- Sternal wound and radial artery wound infections are most commonly caused by Gram +ve organisms, and leg wound infections are caused predominantly by Gram −ve organisms.
- Gram +ve organisms (e.g., *Staphylococcus*, *Streptococcus,* and *Clostridium* spp.) are sensitive to penicillins.
- Gram −ve organisms (pseudomonas, *E. coli*, enterobacter, *Klebsiella*, *Shigella*) are sensitive to ciprofloxacin (500mg bid po: oral absorption better than iv).
- Broad-spectrum antibiotics with good Gram −ve and Gram +ve cover include cefuroxime 750–1.5g iv tid.
- MRSA wound infection should be treated with teicoplanin 400mg iv or im bid for two doses then od, or vancomycin 1g bid, both of which are nephrotoxic.

Wound complications

Sternal instability and sterile dehiscence

Sterile sternal instability and dehiscence is usually due to poor surgical technique, or 'cheese-wiring' of sternal wires through a soft, osteoporotic sternum. The incidence is 1–2%. The patient complains of excessive costochondral pain and 'clicking' on movement, and instability may be palpated when the patient coughs. This may be managed either conservatively with prophylactic antibiotics and support dressings, or with elective sternal rewiring. If it does not become infected, an unstable sternum eventually forms a cartilaginous flexible union or pseudoarthrosis. An unstable sternum is a risk factor for prolonged ventilation and chest infections in the postoperative period.

Sternal dehiscence

Sternal dehiscence presents similarly to abdominal wound dehiscence:

- Serosanguinous discharge from the wound on days 3–5 postoperatively.
- ↑WCC.
- Sudden opening up of wound on coughing, straining.

Get the patient back into bed, reassure them, cover the wound with sterile saline soaked gauze, give the patient 2.5mg im morphine, and put them NPO. Take microbiology swabs. Most patients should be debrided and rewired under general anesthetic. Occasionally, wires have cheese-wired through a soft sternum, e.g., patients are managed conservatively with vacuum dressings and cough-locks.

Hypertrophic and keloid scars

Midline incisions have a tendency to form raised, red hypertrophic scars. Keloid scars (where the lumpy scar tissue exceeds the margin of the scar) are common in patients of African descent. There is no known way of avoiding these unsightly scars, apart from attempting to minimize the length of the sternotomy incision, or considering submammary incisions. Topical silicone gel dressings worn for up to 24h a day for several months will reduce the prominence of hypertrophic scars, but they can relapse when the dressings are abandoned. Keloid scars do not respond to surgical excision, but referral to a plastic surgeon is indicated if the patient is unhappy with the cosmetic result, as steroid injections may offer some improvement.

Sternal wires

Sometimes patients complain of prominent sternal wires. This is commonest in very thin patients. If the sternum is stable, sternal wires may be removed. This should be done under general anesthetic with appropriate invasive monitoring: even a simple removal of sternal wires can turn into an unexpected resternotomy. If the patient is unhappy with just one or two wires make sure you mark them with the patient preoperatively. These can be removed through stab incisions.

Complications of thoracotomy incisions

Post-operative pain is more challenging in thoracotomy patients compared to sternotomy patients (📖 pp648–649), and may progress to chronic post-thoracotomy neuralgia. Wound infections are uncommon and usually respond to appropriate antibiotic therapy. Lung herniation is seen more often with anterior thoracotomies where serratus and lattissimums dorsi do not cover the intercostal spaces which are further apart than they are more posterolaterally: this is diagnosed clinically (pain and protrusion of soft mass on coughing), and confirmed by CT, and can be treated by revision of the incision using a mesh closure. Avoid this by minimizing rib retraction, and a layered closure involving muscle, and circumferential rib sutures.

Hematological complications

Heparin-induced thrombocytopenia (HIT)

HIT occurs in about 5% of patients receiving heparin (5.5% of patients on bovine heparin, 1.0% with porcine heparin). It is characterized by formation of complement-mediated heparin-dependent IgG platelet antibody. It occurs after 5–10 days after initiation of heparin therapy, or after the first dose of heparin in patients with previous exposure to heparin within the last 3 months.

Diagnosis

- Decrease in platelet count by over 30% to $<150 \times 10^9$/L, or by over 50%.
- *And* positive serology for HIT antibodies
- Heparin-induced thrombocytopenia and thrombosis (HITT) occurs in about 20% of patients with HIT and is characterized by major thrombotic episodes, and has a mortality of about 30%.
- Patients may show tachyphylaxis to heparin, as well as bleeding complications; and thrombotic sequelae such as strokes, DVT, PE, peripheral, and splanchnic ischemia.
- Two assays:
  - Screening ELISA assay for antibodies against heparin-platelet factor complex which is more likely to have false positive than
  - Confirmational serotonin release assay (SRA).

Treatment

- Discontinue all heparin therapy, including heparinized saline flushes.
- If it is at all possible, delay any surgery requiring bypass until HIT antibodies are undetectable, and then follow standard heparinization, but do not use heparin in the postoperative period.
- If it is impossible to delay bypass surgery then danaparoid and iloprost are alternatives to heparin with the major disadvantages that they cannot be reversed after bypass, and require specialized assays.
- Hirudin, iloprost, danaparoid, argatroban, and warfarin are alternative anticoagulants to heparin in the postoperative period, and should be used in patients with HITT.

Disseminated intravascular coagulation (DIC)

DIC is extremely rare during bypass but may occur postoperatively as a complication of sepsis, transfusion reaction, drug reaction, transplant rejection, and aortic aneurysm surgery. It is characterized by widespread activation of coagulation, resulting in the formation of intravascular fibrin, fibrin degradation products, consumption platelets and clotting factors, and ultimately thrombotic occlusion of vessels. Patients may present with bleeding from indwelling venous lines, wounds, and minor abrasions.

Diagnosis

There is no single diagnostic test. The following findings suggest DIC:

- Sudden fall in platelet count to $<100 \times 10^9$/L.
- Bleeding and/or thrombotic complications.
- ↑APTT, PT, INR.
- ↑fibrin degradation products.
- ↓fibrinogen in severe DIC.

Management

The key is to treat the underlying disorder. Bleeding patients should receive FFP, platelets, blood, and cryoprecipitate as indicated by coagulation screens. Patients with thrombosis should be heparinized.

Excessive heparinization

Management depends on whether the patient is bleeding (see 📖 pp145, 150). In the stable patient, who is not bleeding, simply stop the heparin infusion and recheck APTT in 4-6 hours. Heparin may be reversed rapidly with protamine (25–50mg iv) or FFP. The complications of protamine are described on 📖 pp116–117; the complications of transfusion are described on 📖 pp148–149.

Excessive warfarinization

Management depends on whether the patient is bleeding, and why they are warfarinized:

- If the INR is <6.0 and the patient is not bleeding simply omit warfarin and recheck INR daily, restarting warfarin once the INR is within target range. Check for evidence of liver dysfunction (📖 p276), and common contributing factors such as right heart failure (📖 p191), and drug interactions which elevate INR in patients on warfarin (e.g., amiodarone, cephalosporins, ciprofloxacin, tylenol, omeprazole).
- Check medications for interactions that may be causing INR to increase, e.g., amiodarone, antibiotics
- If the INR is >6.0 give 10mg vit K im.
- If the patient is bleeding give FFP (📖 p145).
- If the patient has a mechanical valve the risk of thromboembolic events when anticoagulation is reversed is about 0.02% per day: anticoagulation with heparin or warfarin should be recommenced within 1–2 weeks once any bleeding complications have resolved.

Hemolysis

Transient hemolysis manifesting as mild hematuria on arrival in ICU is most commonly due to cardiopulmonary bypass (📖 p95, p103, p104). Persistent or late hemolysis may be due to small, high velocity paravalvular leaks (📖 p266), is not uncommon in VAD patients (📖 p499), and may be a manifestation of transfusion reactions (📖 p148) or drug toxicity.

Hemolysis is diagnosed by increased presence of reticulocytes, hemoglobinuria, and frequently raised LFTs, and bilirubin. Management is supportive: keep the patient hydrated and avoid nephrotoxic and hepatotoxic agents to reduce the risk of renal and hepatic compromise, transfuse if indicated. Severe, refractory cases due to paravalvular leaks may require redo surgery.

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